

"Write here, write now": Spelling difficulty disrupts parallel planning in sentence production

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To what extent does difficulty with word spelling affect our ability to convey ideas in writing? This research aims to ...

- ▶ establish when during sentence production word features are being retrieved.
- ▶ evaluate how spelling difficulty affects our ability to plan language.

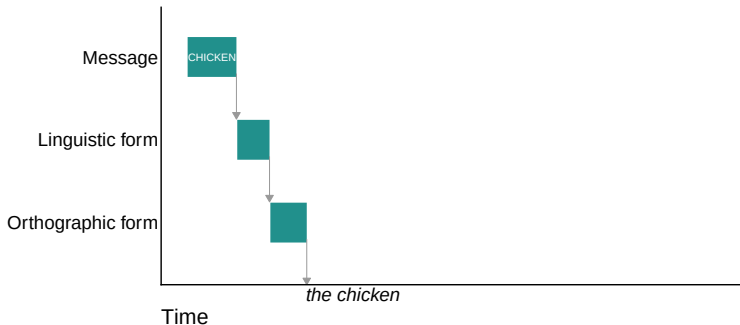
How are levels of language planning coordinated in real time?

Planning language takes time!



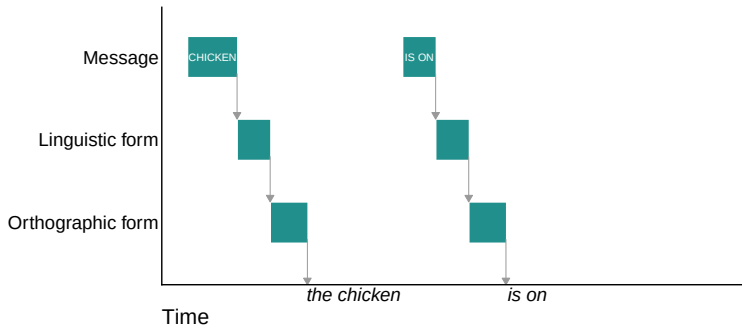
Serial view: Speaker / writers pause at linguistic edges to plan upcoming information.

Planning language takes time!



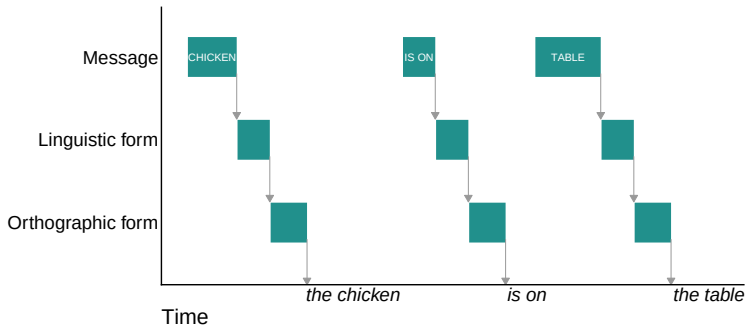
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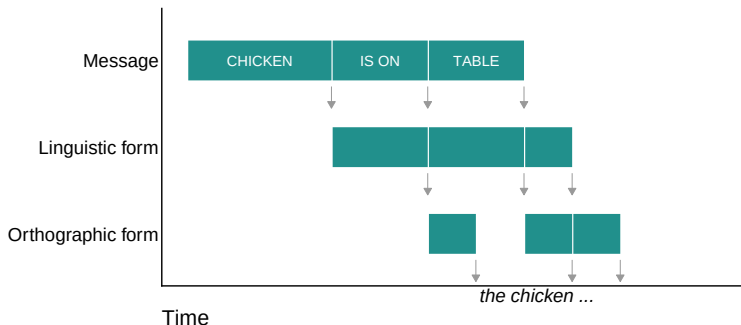
Planning language takes time!



Parallel view: Planning unfolds in parallel to output (Olive, [2014](#); Roeser et al., [2025](#); Van Galen, [1991](#)).

Pauses during production result from delays in information flow! Slow orthographic retrieval delays upstream processes.

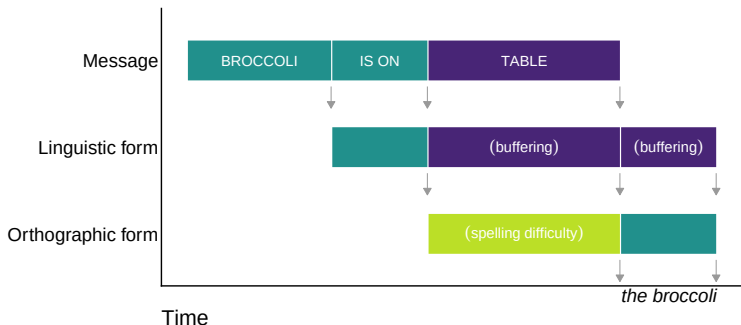
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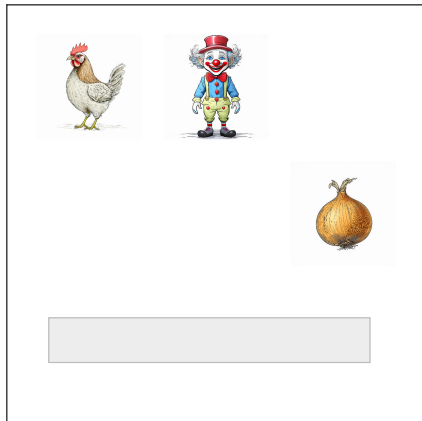
Planning language takes time!



Parallel view: Planning unfolds in parallel to output (Olive, 2014; Roeser et al., 2025; Van Galen, 1991).

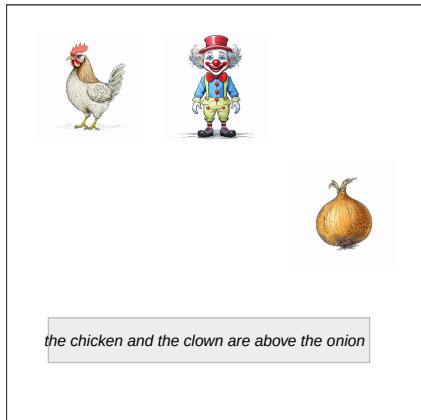
Pauses during production result from delays in information flow! Slow orthographic retrieval delays upstream processes.

Describe the arrangement from left to right!



Example screen (Roeser et al., 2019; Smith & Wheeldon, 1999)

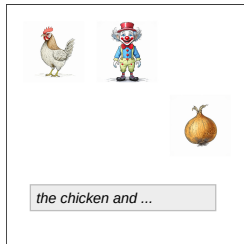
Describe the arrangement from left to right!



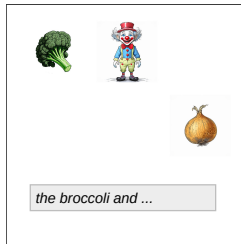
Example screen (Roeser et al., [2019](#); Smith & Wheeldon, [1999](#))

Design: Example item set

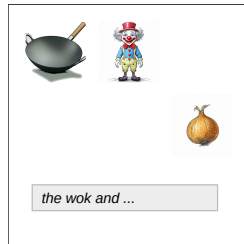
N1: easy spelling / long



N1 difficult spelling / long



N1: easy spelling / short



Prediction: Typing hesitations for N2 (*clown*) result from (1) when N1 spelling (*broccoli*) couldn't be retrieved in time and (2) when N1 length (*wok*) didn't allow for parallel planning.

- ▶ 36 items counterbalanced for prime structure (simple, conjoined subject) and N1 word type.
- ▶ 24 fillers with structures different from items.
- ▶ N1 word type manipulation: easy spelling (long), easy spelling (short), difficult spelling (long):
 - ▶ Long: ≥ 2 syllables
 - ▶ Difficult: spelling diversity $H > 0.5$

What is “spelling diversity”?

Write the name of what you see!

What is “spelling diversity”?



What is “spelling diversity”?

Example summary for “raccoon”(N = 97 participants).

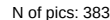
response	N_{spell}	N_{name}	Pr_{spell}	H_{spell}
raccoon	41	75	.55	1.31
racoon	30	75	.40	1.31
raccon	2	75	.03	1.31
raccoo	1	75	.01	1.31
racoo	1	75	.01	1.31

Note. H = diversity statistic; Pr = proportion of responses.

Spelling diversity H_{spell} represents the frequency of a spelling used relative to the number different spellings (Lachman, 1973; Torrance et al., 2018).

$$H_{spell} = \sum_{i=1}^K p_i \times \log_2\left(\frac{1}{p_i}\right), \quad (1)$$

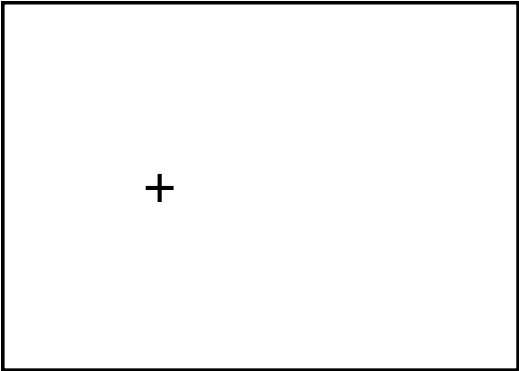
where k is the number of spellings for a picture name; p_i is the proportion of participants producing the i^{th} spelling.



Ten ppts per counterbalancing list. After data cleaning:

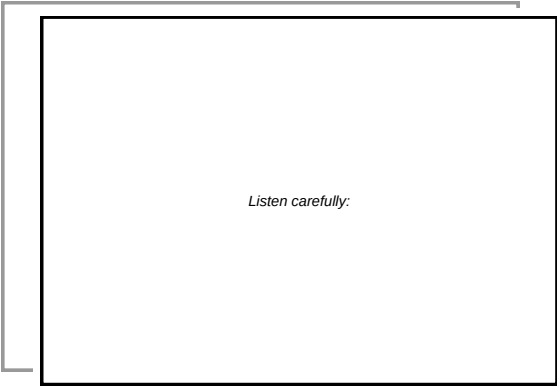
- ▶ Experiment 1: 56 ppts (6 lists)
- ▶ Experiment 2: 80 ppts (9 lists)
- ▶ Experiment 3: 96 ppts (12 lists)

Trial sequence



+

Trial sequence



Listen carefully:

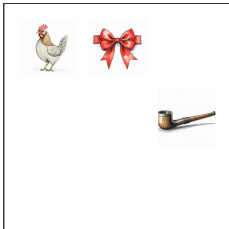
Trial sequence

"The chicken and the bow are above the pipe."

Listen carefully:

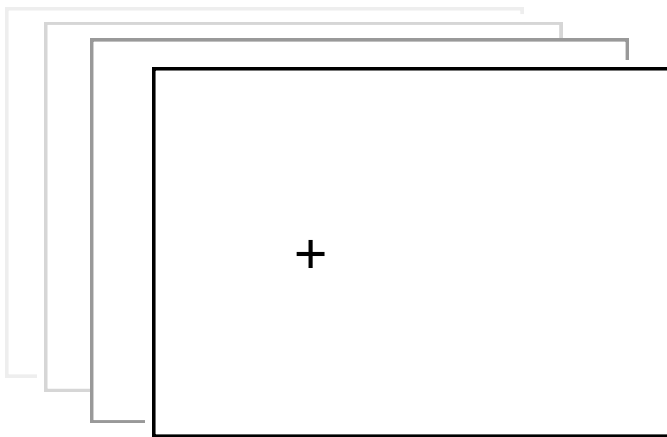
Trial sequence

Does the image on the screen match what you heard?



Press Y (Yes) or N (No)

Trial sequence



Trial sequence

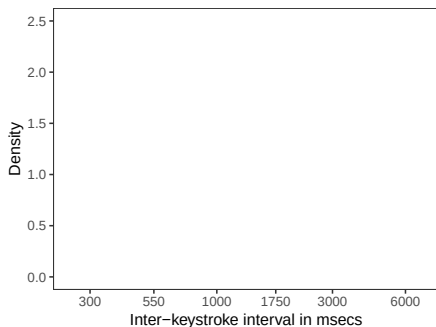


Trial sequence

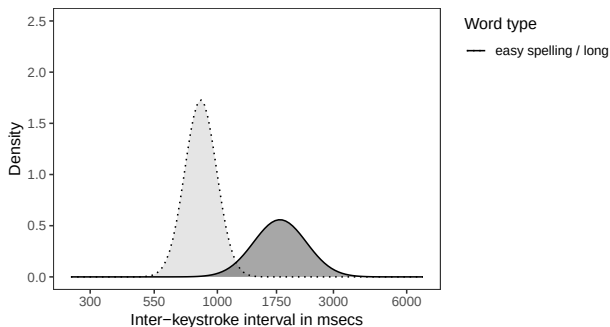


the chicken and ...

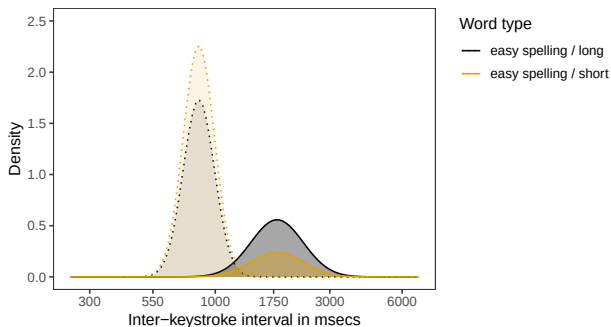
Data analysis: typing as mixture process



Inter-keystroke intervals were modelled as Bayesian mixed-effects mixture model (Roeser et al., 2024, 2025). For a tutorial see: <https://rpubs.com/jensroes/mixture-models-tutorial>.

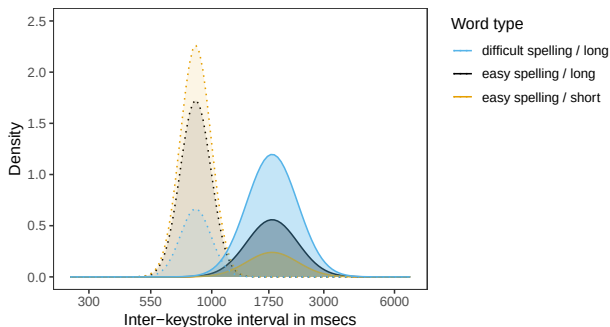


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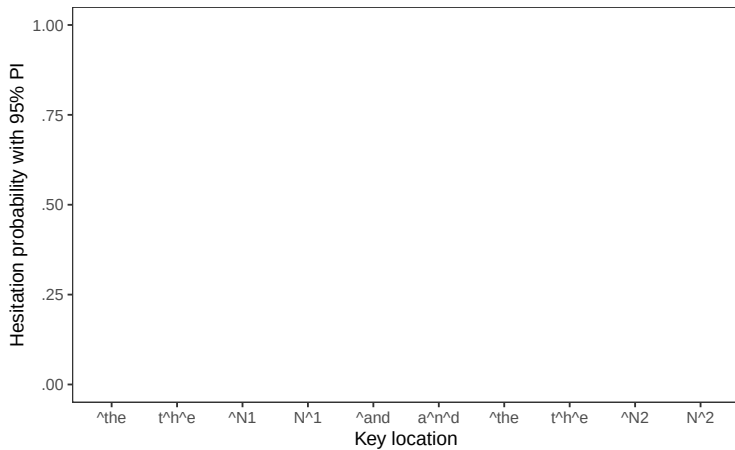
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Data analysis: typing as mixture process

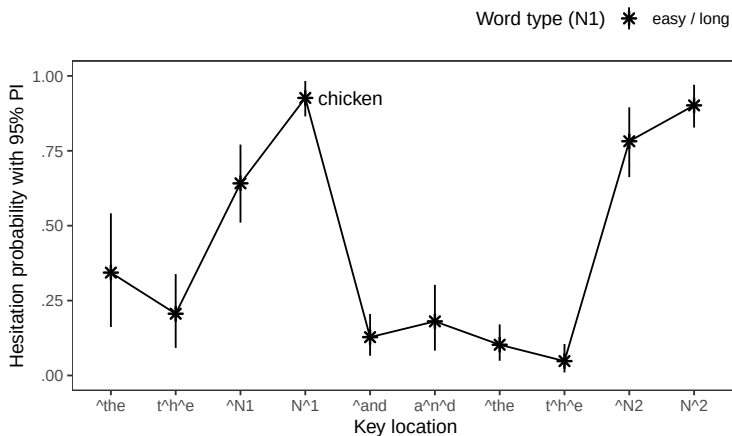


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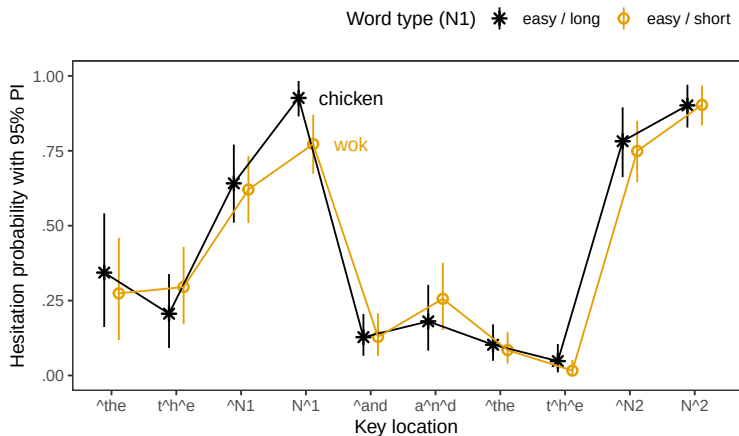
Experiment 1: typing timecourse



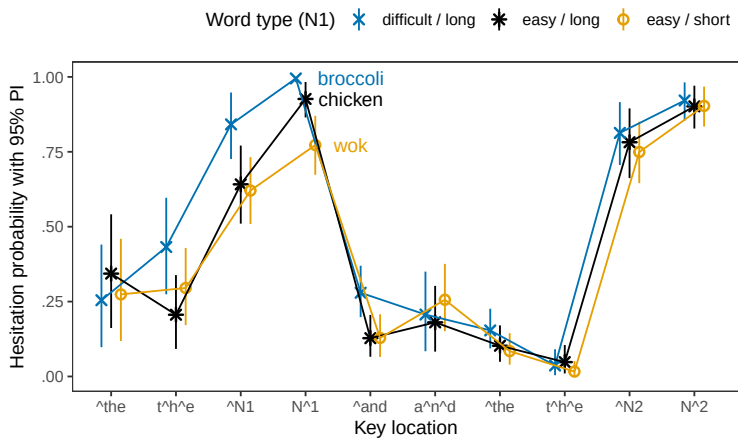
Experiment 1: typing timecourse



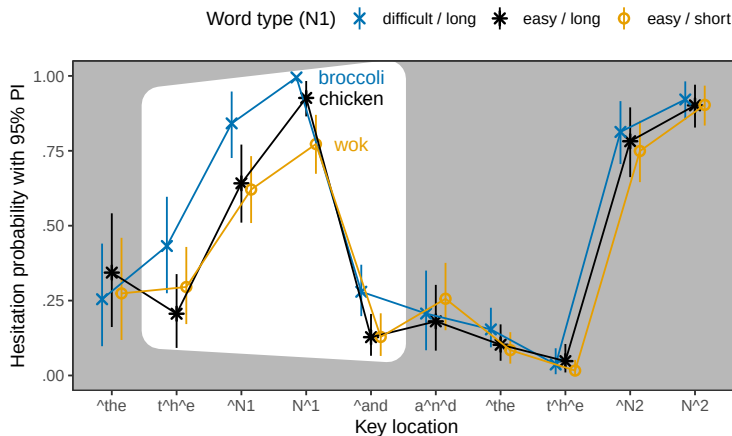
Experiment 1: typing timecourse



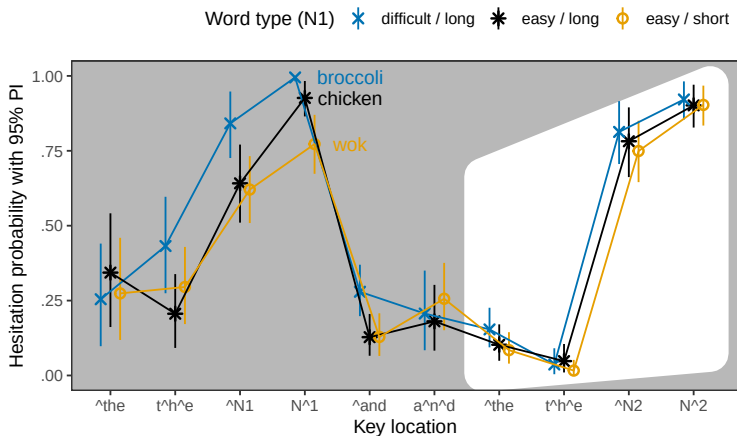
Experiment 1: typing timecourse



Experiment 1: typing timecourse



Experiment 1: typing timecourse



Experiment 2

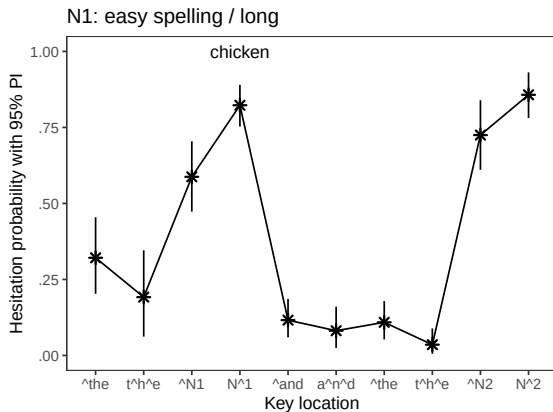
Does typing and the buy writers enough time to avoid hesitations before N2?

Additional prime condition; two target structures:

- ▶ With N2 determiner: *The N1 and the N2 are above the N3*
- ▶ Without N2 determiner: *The N1 and N2 are above the N3*

Prediction: If the distance between nouns constrains parallel planning, we expect more hesitations before N2 when *the* was omitted.

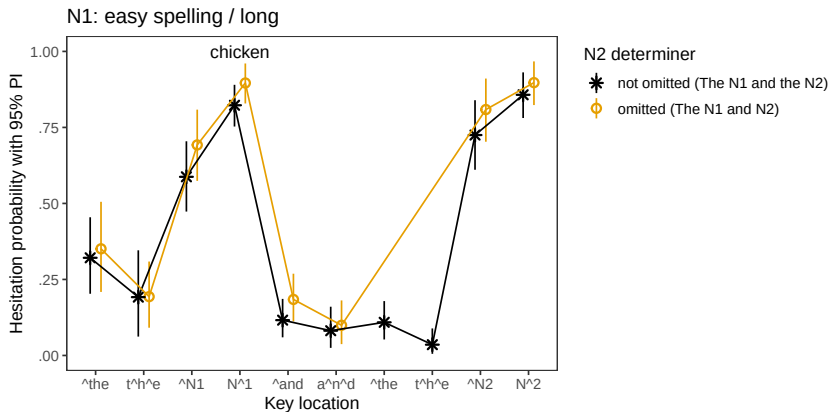
Experiment 2: typing timecourse



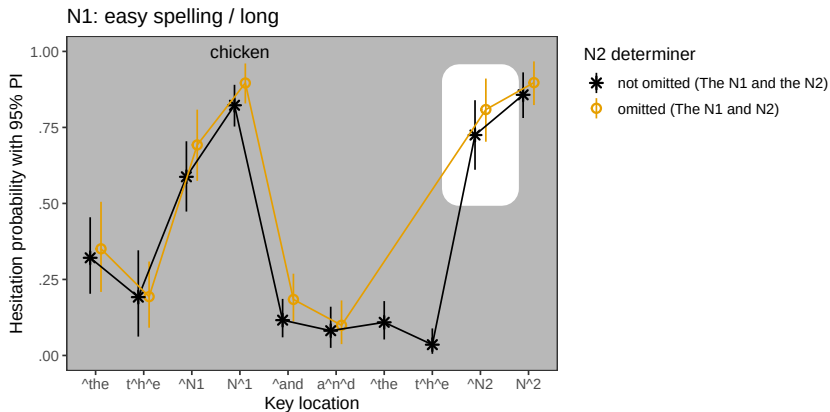
N2 determiner

* not omitted (The N1 and the N2)

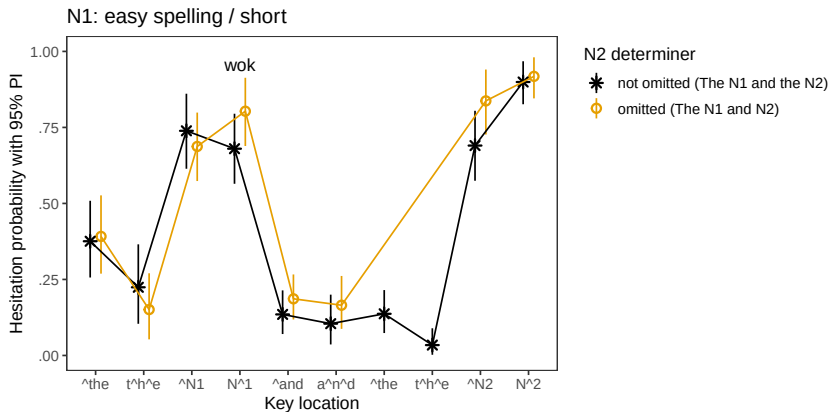
Experiment 2: typing timecourse



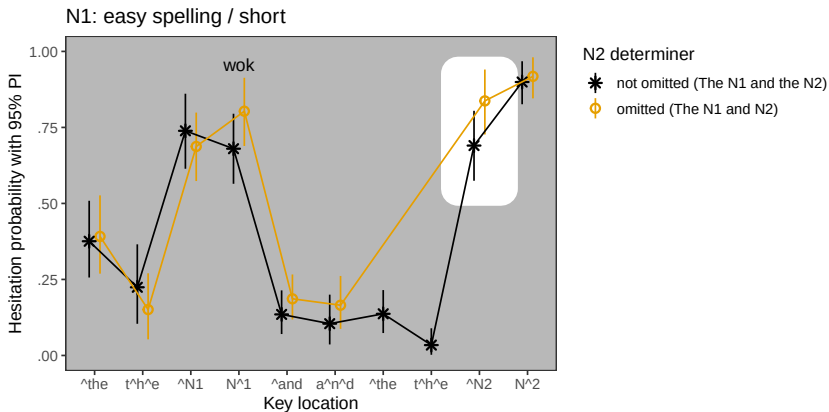
Experiment 2: typing timecourse



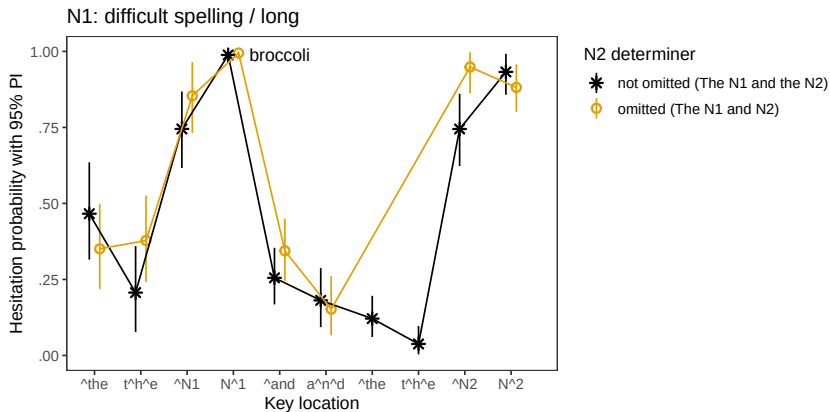
Experiment 2: typing timecourse



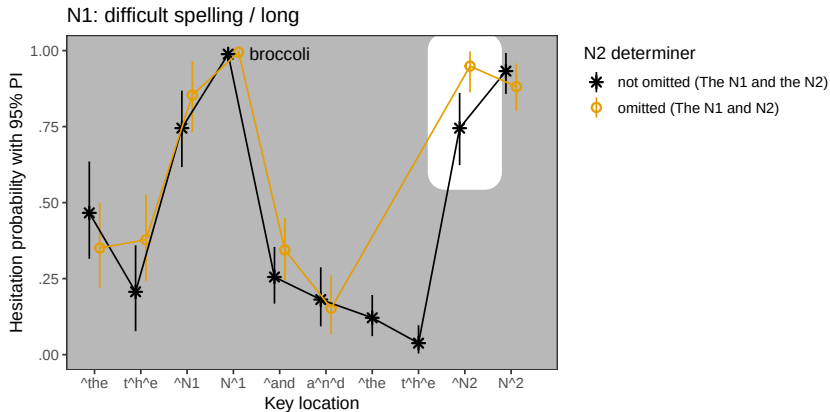
Experiment 2: typing timecourse



Experiment 2: typing timecourse



Experiment 2: typing timecourse



Experiment 3

Are hesitations related to delayed N2 retrieval?

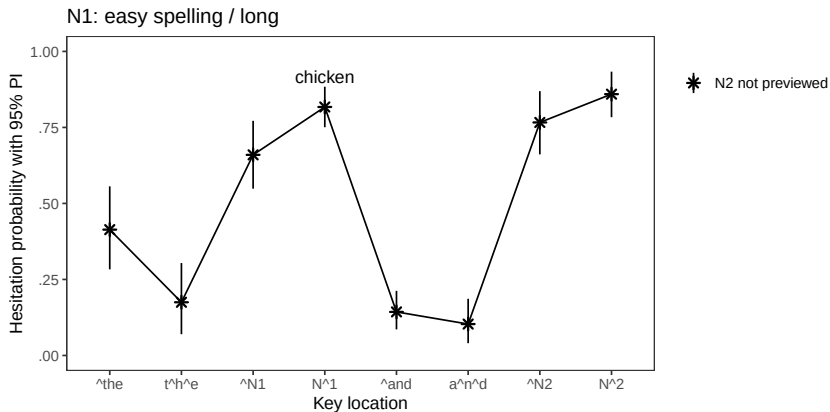
Prime with (auditory) N2 preview:

- ▶ Previewed: *The N1 and clown are above the pipe*
- ▶ Not previewed: *The N1 and bow are above the pipe*

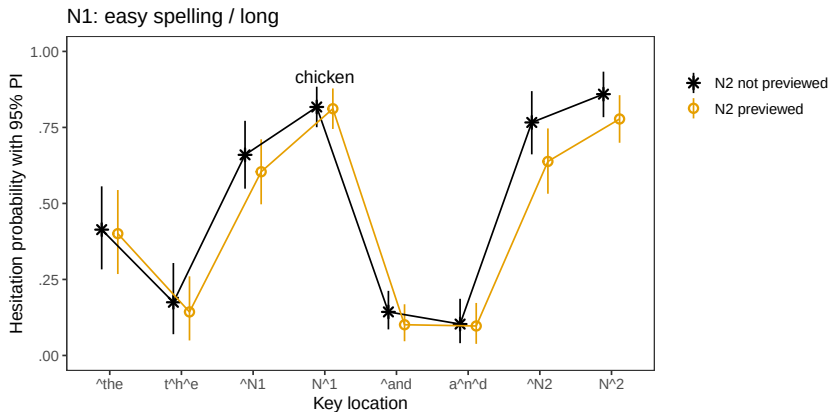
Target: *The N1 and clown are above the onion*

Prediction: If the planning bottleneck delayed N2 retrieval, preview will reduce associated hesitations.

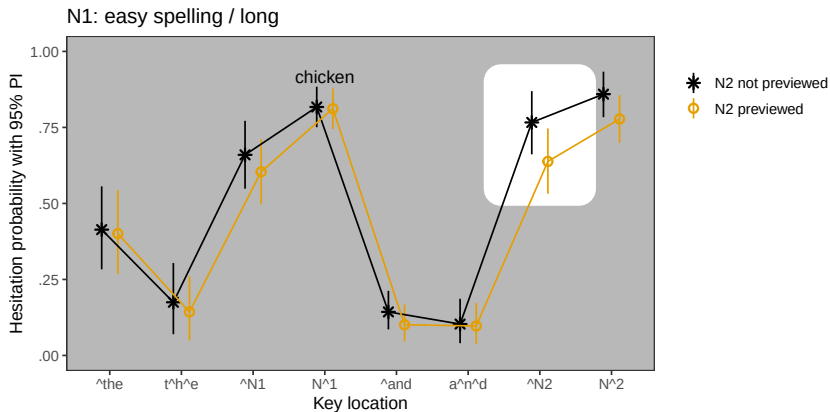
Experiment 3: typing timecourse



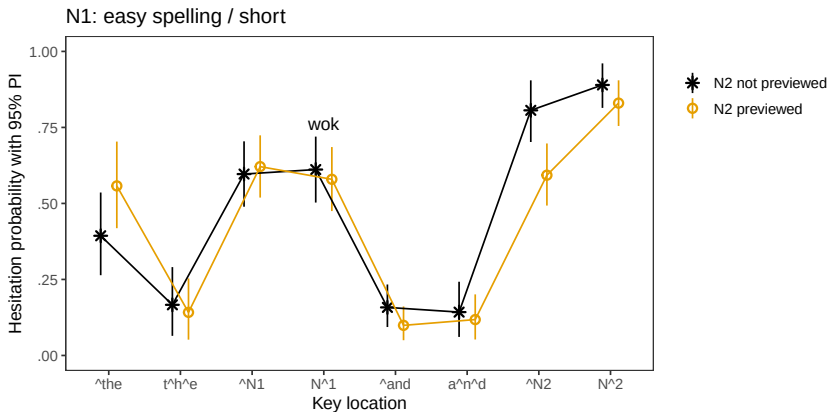
Experiment 3: typing timecourse



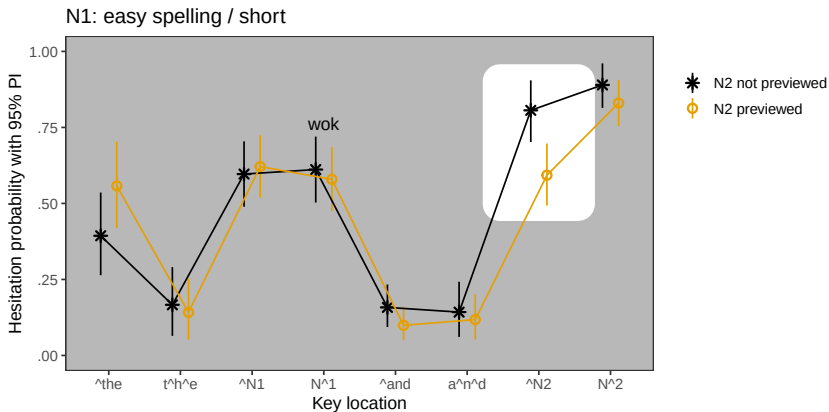
Experiment 3: typing timecourse



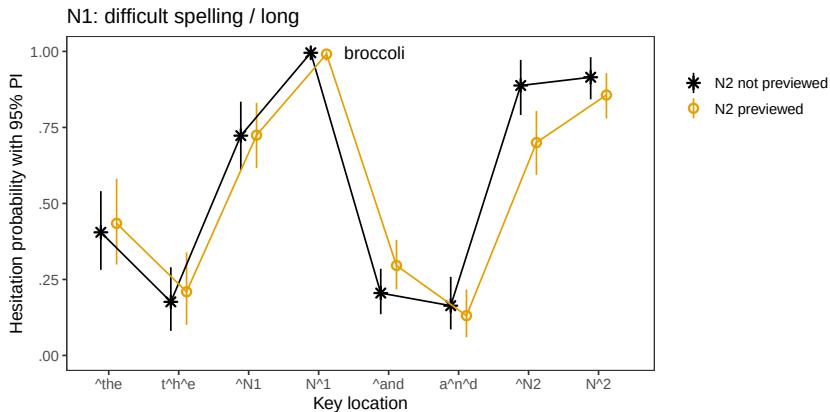
Experiment 3: typing timecourse



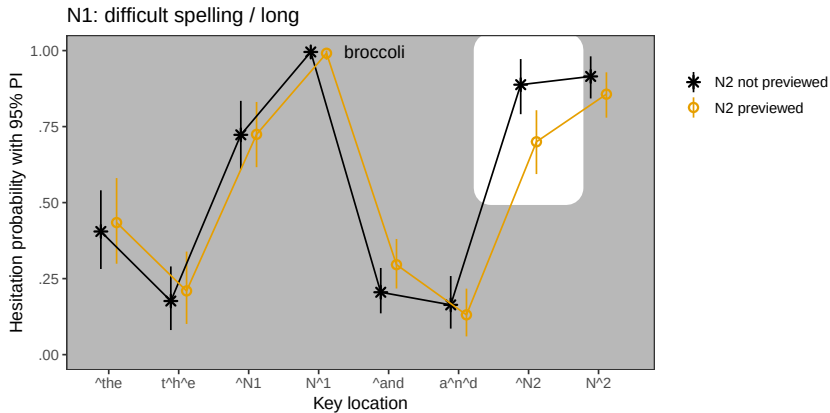
Experiment 3: typing timecourse



Experiment 3: typing timecourse



Experiment 3: typing timecourse



- ▶ Hesitations related to N1 spelling affect N2 retrieval when there was not enough time between N1 and n2 for parallel lexical retrieval of N2.
- ▶ Long and / or functional words – e.g. *and the* – provide time for parallel lexical retrieval (see also Griffin, [2003](#)).
- ▶ Written sentence planning unfolds in parallel to production: difficulty with spelling creates a planning bottleneck with knock-on effects on higher level planning.
- ▶ Serial models of language planning cannot explain these results (Flower & Hayes, [1980](#); Hayes, [2012](#)).

Thanks for listening!

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- Griffin, Z. M. (2003). A reversed word length effect in coordinating the preparation and articulation of words in speaking. *Psychonomic Bulletin & Review*, 10(3), 603–609.
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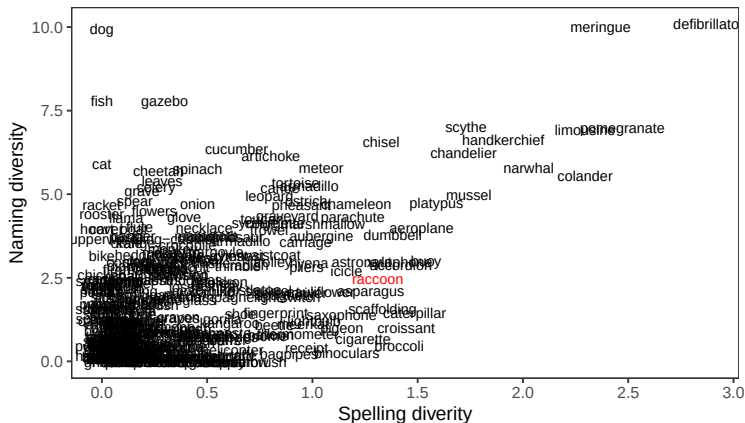
References II

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References III

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Spelling and naming diversity



N of pics: 383

Typing as mixture process

Inter-keystroke intervals were analysed in mixture models with shifted-lognormal distributions (Roeser et al., [2024](#), [2025](#)).

$$(\log(iki_{ij}) - shift_i) \sim \theta_{condition} \times \mathcal{N}(\beta_{location} + \delta_{location} + u_i + w_j, \sigma_{e'}^2) + \\ (1 - \theta_{condition}) \times \mathcal{N}(\beta_{location} + u_i + w_j, \sigma_e^2)$$

where:

$$u_i \sim \mathcal{N}(0, \sigma_u^2)$$

$$w_j \sim \mathcal{N}(0, \sigma_w^2)$$

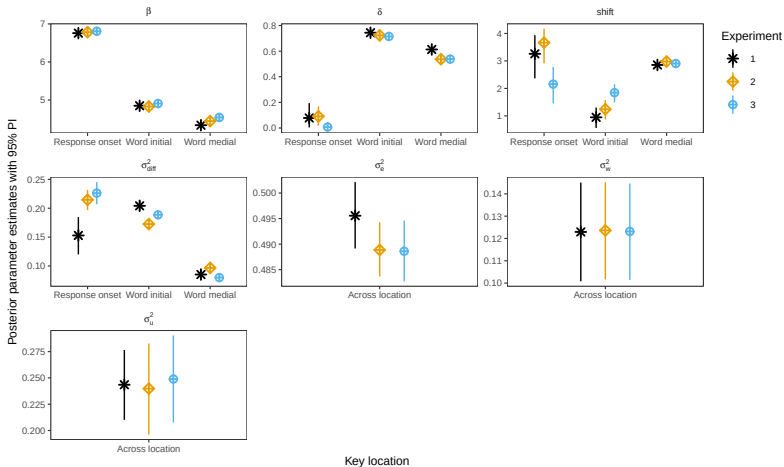
constraints:

$$\delta, \text{ (and all } \sigma\text{s)} > 0$$

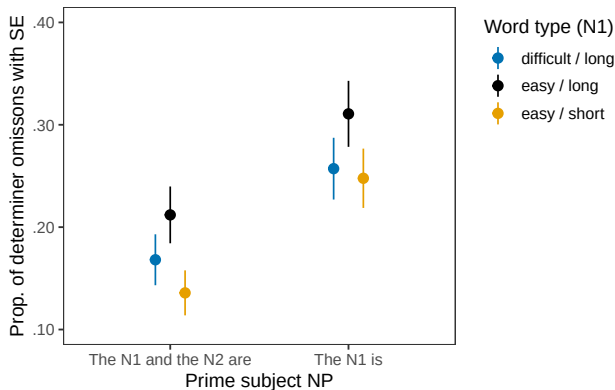
$$\sigma_{e'}^2 > \sigma_e^2$$

θ returns the proportion of hesitant inter-keystroke intervals for each key-location, prime and N1 condition.

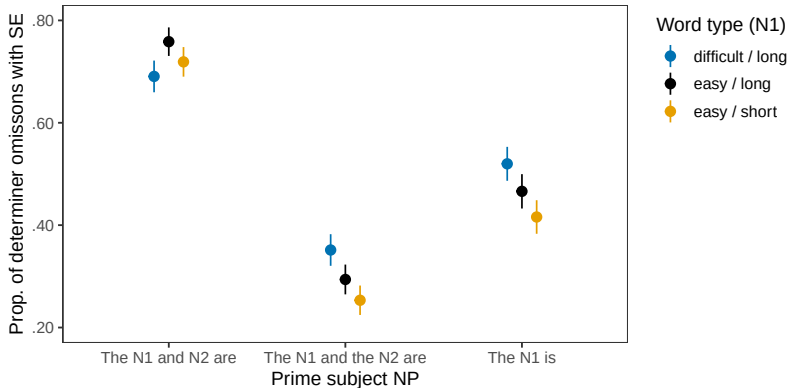
Most other mixture model parameter estimates



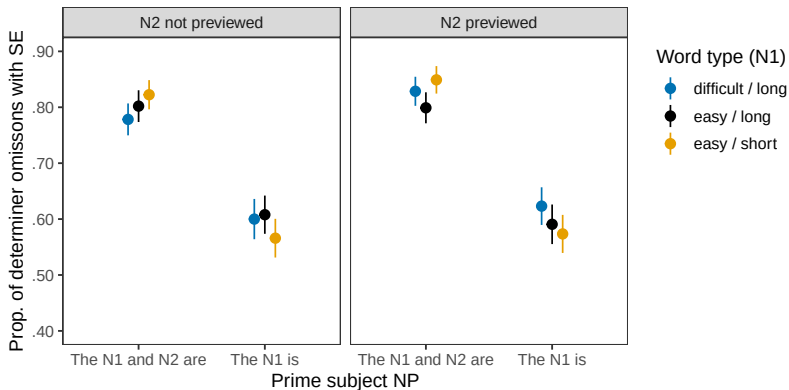
Experiment 1: determiner omissions before N2



Experiment 2: determiner omissions before N2



Experiment 3: determiner omissions before N2



Experiment 3: N2-initial hesitations

